

Lecture 11

Random and Mixed Effects Models, Computer Experiments

Reading: Oehlert 2010 Chapter 11.1-11.7; DAE 2017 Chapters 17.1-17.3; 17.7& 20

DSA 8020 Statistical Methods II

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Notes

Agenda

1 Random and Mixed Effects Models

2 Computer Experiments



Notes

Fixed Effects

Most settings we have dealt with so far (except the analysis of split-plot designs) have involved **fixed effects**:

CRD:	$y_{ij} = \mu + \alpha_i + \epsilon_{ij}$
RCBD:	$y_{ij} = \mu + \alpha_i + \beta_j + \epsilon_{ij}$
Factorial:	$y_{ij} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ij}$

- The treatment effects are unknown but constants $\Rightarrow$ if we ran the experiment over again, would expect the same treatment effects
- We can increase the power of all of our tests by increasing the sample size n
- We perform inference on the treatment effects via t -tests and F -tests



Notes

Random Effects

Random effects models look very similar to fixed effects models. For example, we could have

y_{ij} = \mu + \alpha_i + \epsilon_{ij}.

The difference is in the assumptions we make for the treatment effects

Fixed Effects	Random Effects
Treatment effects \alpha_i s are unknown constants that add to zero (or some other constraint)	\alpha_i s \sim N(0, \sigma_\alpha^2) \alpha_i s are independent of \epsilon_{ij}



Notes

How and Why Are Things So Different?

Fixed effects:

- The treatments are the treatments and they are unchanging
- If we rerun the experiment, we are still studying the same treatments

Random effects:

- The treatments are a random sample from a population of potential treatments
- If we rerun the experiment, we are looking at an entirely new sample of treatments
- Inference is on the population of potential treatments



Notes

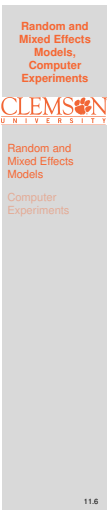
Variance Components

Fixed effects:

- Var(y_{ij}) = \sigma^2
- All y_{ij} s are independent of each other
- Interest is about \alpha_i s

Random effects:

- Var(y_{ij}) = \sigma_\alpha^2 + \sigma^2
- Cor(y_{ij}, y_{kl}) = \begin{cases} 0 & \text{if } i \neq k \\ \frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma^2} & \text{if } i = k; j \neq l \\ 1 & \text{if } i = k; j = l \end{cases}
- Interest is (mostly) about \sigma_\alpha^2



Notes

An Example of Fixed Effects vs Random Effects

- 1 Compare reading ability of 10 2nd grade classes in NY:

Select $g = 10$ specific classes of interest. Randomly choose n students from each classroom. Want to compare $\alpha_i s$ (class-specific effects) $\Rightarrow$ Fixed effects
- 2 Compare variability among all 2nd grade classes in NY:

Randomly choose $g = 10$ classes from large number of classes. Randomly choose n students from each classroom. Want to assess σ^2_{α} (class to class variability) $\Rightarrow$ Random effects

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Random Effects Model (CRD)

$y_{ij} = \mu + \alpha_i + \epsilon_{ij},$

- where
- μ is the overall mean
 - α_i : ith treatment effect and $\alpha_i \sim N(0, \sigma^2_{\alpha})$
 - $\{\alpha_i\}$ and $\{\epsilon_{ij}\}$ independent
 - The hypotheses are:

$H_0 : \sigma^2_{\alpha} = 0$
 $H_a : \sigma^2_{\alpha} > 0$

One can use either “old school” method (ANOVA) or “new school” method (REML) to make inference about σ^2_{α}

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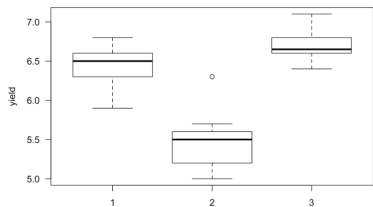
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Random Effects Example

Suppose that an agronomist is studying a large number of varieties of soybeans for yield. The agronomist randomly selects three varieties, and then randomly assigns each of those varieties to 10 of the 30 available plots

Soybean	Yield
V1	6.6, 6.4, 5.9, 6.6, 6.2, 6.7, 6.3, 6.5, 6.5, 6.8
V2	5.6, 5.2, 5.3, 5.1, 5.7, 5.6, 5.6, 6.3, 5.0, 5.4
V3	6.9, 7.1, 6.4, 6.7, 6.5, 6.6, 6.6, 6.6, 6.8, 6.8



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Fixed Effects Analysis

```
> fixef <- lm(yield ~ var)
> anova(fixef)
Analysis of Variance Table

Response: yield
      Df Sum Sq Mean Sq F value    Pr(>F)
var       2  8.306   4.1530  49.593 9.114e-10 ***
Residuals 27  2.261   0.0837
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> coefficients(fixef)
(Intercept)      var2      var3
      6.45      -0.97       0.25
```

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Random Effects Analysis

```
> library(lme4)
> randef <- lmer(yield ~ 1 + (1|var), REML = TRUE)
> summary(mod1)
Linear mixed model fit by maximum likelihood . t-tests
use Satterthwaite's method [lmerModLmerTest]
Formula: yield ~ 1 + (1 | var)

      AIC      BIC    logLik deviance df.resid
 27.2    31.4    -10.6     21.2        27

Scaled residuals:
    Min       1Q   Median       3Q      Max
-1.8755 -0.6033  0.1245  0.5068  2.7574

Random effects:
Groups   Name      Variance Std.Dev.
var      (Intercept) 0.26849   0.5182
Residual              0.08374   0.2894
Number of obs: 30, groups: var, 3

Fixed effects:
              Estimate Std. Error    df t value Pr(>|t|)
(Intercept)    6.2100     0.3038 3.0000  20.44 0.000256
```

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Concrete Cylinder Example Revisited

Suppose you are manufacturing concrete cylinders for bridge supports. There are three ways of drying concrete (say A, B, and C), and you want to find the one that gives you the best compressive strength. The concrete is mixed in batches that are large enough to produce exactly three cylinders, and your production engineer believes that there is substantial variation in the quality of the concrete from batch to batch.

Treatment	Batch					Trt Sum
	1	2	3	4	5	
A	52	47	44	51	42	236
B	60	55	49	52	43	259
C	56	48	45	44	38	231
Batch Mean	168	150	138	147	123	726

If we were treat the batch effects as random effects, then we have a [Mixed Effects Model](#)

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Concrete Cylinder Example: Mixed Effects Analysis

```
> randef <- lmer(x ~ trt + (1|blk), REML = TRUE, data = dat)
> summary(randef)
Linear mixed model fit by REML. t-tests use
Satterthwaite's method [lmerModLmerTest]
Formula: x ~ trt + (1 | blk)
Data: dat

REML criterion at convergence: 71.1

Scaled residuals:
    Min       1Q   Median       3Q      Max
-1.1417 -0.6147 -0.1494  0.5772  1.3390

Random effects:
Groups   Name              Variance Std.Dev.
blk      (Intercept) 28.35      5.324
Residual                    5.85      2.419
Number of obs: 15, groups: blk, 5

Fixed effects:
              Estimate Std. Error    df t value Pr(>|t|)
(Intercept)  47.200      2.615   5.054  18.047 8.76e-06
trtB         4.600      1.530   8.000   3.007  0.0169
trtC        -1.000      1.530   8.000  -0.654  0.5316
```

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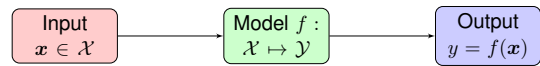
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What is a Computer Experiment

In some situations, it is economically, ethically, or simply not possible to run a **physical experiment**. Instead, the following scenario might be feasible:

- the physical process can be described by a mathematical model (e.g., a system of differential equations)
- computer code (simulator) can be written to compute the response from the mathematical model



In this case, a researcher can conduct a **computer experiment** by running the computer code, which serves as a proxy for the physical process, to compute a “response” at any combination of values of the inputs

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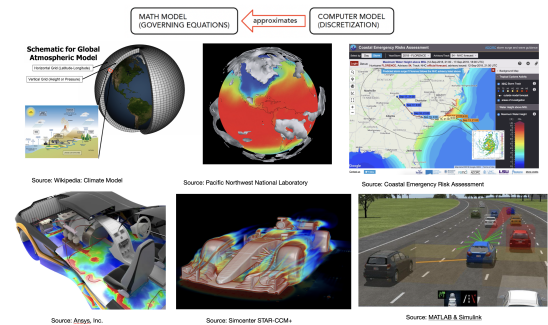
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Examples of Computer Models



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Computer Experiments vs. Physical Experiments

- “Experimental results are believed by everyone, except for the person who ran the experiment”
- “Computational results are believed by no one, except the person who wrote the code”

Replication, randomization and blocking are irrelevant for a computer experiment because many **computer codes are deterministic** and **all the inputs to the code are known and can be controlled**

Here we are concerning about **design** and **analysis** of computer experiments:

- **Design**: Which configurations of $\{x_i\}_{i=1}^n$ to run the computer model
- **Analysis**: How to estimate the input-output relationship $y = f(x)$ using data $\{x_i, y_i\}_{i=1}^n$ from a computer experiment

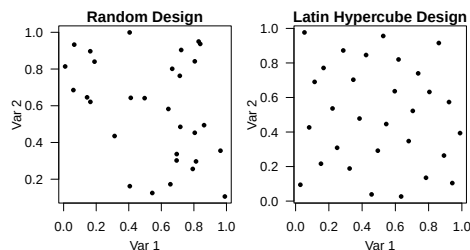


Notes

Design of Computer Experiments

Question: where to make the runs, i.e., the selection of inputs $\{x_i\}_{i=1}^n$ for a given computational budget n .

Example: $x_i = (x_{i1}, x_{i2})^T \in [0, 1]^2$ with $n = 30$



The **Latin hypercube design** is an example of **space-filling designs**, which allow for an evenly spread of points encompassing the entire design space.



Notes

Analysis of Computer Experiments (aka Emulation)

Goal: fit a statistical model to the computer model inputs-output $\{y_i, x_i\}_{i=1}^n$ to “emulate” the simulator and to quantify the prediction uncertainty for $y(x_{\text{new}})$ via a **Gaussian Process Model** $\text{GP}(m(\cdot), K(\cdot, \cdot))$, where

- $m(x) = E[y(x)]$ is the **mean function**, usually takes a simple form, e.g., $m(x) = \mu$
- $K(x, x') = \text{Cov}(y(x), y(x'))$ is the **covariance function**, usually parametrized by “distance”. e.g., $K(x, x') = C(x, x'; \theta) = \sigma^2 \prod_{j=1}^p C_j(d(x_j, x'_j); \theta_j)$.

Parameters $(\mu, \sigma^2, \{\theta_j\}_{j=1}^p)$ can be estimated by fitting a GP model to $\{y_i, x_i\}_{i=1}^n$ via **maximum likelihood method**. The prediction (e.g., predicting $y(x_{\text{new}})$) and prediction uncertainty can be carried out using the Gaussian conditional distribution formula

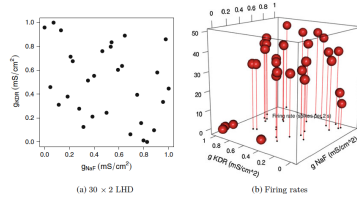


Notes

Neuron Experiment [pp.776-778, DAE 2017]

The firing rate of a neuron at +380 pA current injection of a young monkey is modeled as a deterministic function of two input variables:

- x_1 g_{NaF} : maximal conductance of the transient sodium
- x_2 g_{KDR} : maximal conductance of the delayed-rectifier potassium



Source: Fig. 20.6, DAE 2017

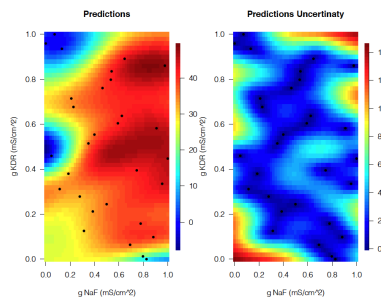
The goal is to reconstruct the 2D surface within the input space



Notes

Neuron Experiment Result

- A GP with squared exponential covariance function (i.e., $C(x, x') = \sigma^2 e^{-[\theta_1(x_1 - x'_1)^2 + \theta_2(x_2 - x'_2)^2]}$) is fitted to $\{y_i, x_i\}_{i=1}^n$ with the estimated parameters $\hat{\mu} = 27.61$, $\hat{\sigma}^2 = 251.86$, $\hat{\theta}_{NaF} = 5.03$, $\hat{\theta}_{KDR} = 50.22$.
- With these estimated parameters one can calculate the predictions (**Left**) and their prediction uncertainties (**Right**)



Notes

Summary

These slides cover:

- Random and Mixed Effects Models
- Computer Experiments: Concepts, Design and Analysis

R functions to know:

- **Random and Mixed Effects Modeling:** `lmer` from the packages `lme4`/`lmerTest`
- **Design and Analysis of Computer Experiments:** `maximinLHS` from the package `lhs` for conducting Latin hypercube sampling designs and `mlegp` from the package of the same name for GP emulation



Notes